Hadith Hallucination Benchmark

A benchmark pipeline that measures how much local Large Language Models (LLMs), run via Ollama, **hallucinate** when asked to recall the exact text of a specific Hadith given only its reference (Book, Chapter, Hadith Number) — instead of quoting the real text, do they confidently invent one?

Prepared for: Muslim-in-ML Hadith project | Dataset: 6 major Hadith collections | Models covered in this version: mistral:7b, deepseek-r1:8b, llama3.1:8b

1. Why This Project Exists

Hadith texts are precise, sourced religious records with a defined chain of narration (isnad) and an exact wording. If a language model is asked “what does Hadith #442 of Sunan Abi Da'ud say?” and it does not actually know the answer, the concerning failure mode is not silence — it is a model that **confidently fabricates a plausible-sounding hadith**, complete with a fake narrator and a fake source citation, rather than admitting uncertainty. This is a serious risk in a religious-knowledge context, where a fluent, well-formatted fabrication can be mistaken for a real narration.

This benchmark quantifies that failure mode at scale: across multiple LLMs, across two languages (English and Arabic), and across all six major Hadith collections.

2. Dataset

The dataset (`dataset/all_hadiths_clean.csv`) contains hadiths from six books, each with an Arabic and English text field:

- id, hadith_id — identifiers
- source — the book name (Sahih Bukhari, Sahih Muslim, Sunan Abi Da'ud, Jami' al-Tirmidhi, Sunan an-Nasa'i, Sunan Ibn Majah)
- chapter_no, chapter — chapter number and name
- hadith_no — hadith number within the book
- chain_indx — narrator chain index
- text_ar, text_en — the ground-truth Arabic and English hadith text

3. Project Structure

```
hadith_benchmark/
  dataset/all_hadiths_clean.csv — raw source data
  data/sampled_1000_per_book.csv — stratified sample, 1000/book, seed=42
  prompts/prompt_english.txt, prompt_arabic.txt
  results/*.csv — raw + evaluated LLM outputs, final_summary.csv
  doc/Hadith_Hallucination_Benchmark.pdf — this document
  01_sample_data.py, 02_run_llm.py, 03_evaluate.py, 04_results_summary.py
  requirements.txt, README.md
```

4. Pipeline Walkthrough

Step 1 — 01_sample_data.py

Reads the raw CSV, strips whitespace from the `source` column, and draws a **stratified random sample of 1000 hadiths per book** (reproducible with `random_seed=42`). Saves to `data/sampled_1000_per_book.csv` and prints per-book counts as a sanity check.

Step 2 — 02_run_llm.py

For a chosen `--model` (e.g. mistral:7b) and `--lang` (english or arabic):

- Loads the matching prompt template from prompts/.
- For each sampled hadith, fills in the source, chapter number, chapter name, and hadith number into the prompt — the model receives **only the reference, never the actual text**.
- Sends the filled prompt to Ollama's local API at http://localhost:11434.
- Records the model's raw response next to the real ground-truth text.
- Uses a tqdm progress bar, a 0.5 second delay between calls, and writes "ERROR" for any failed call instead of crashing the run.

Output columns: hadith_id, source, chapter_no, hadith_no, chapter, ground_truth_en, ground_truth_ar, llm_response, model, lang.

Prompt design note: the prompt does not forbid the model from including the chain of narrators (isnad), since real ground-truth text often begins with narrator attribution (e.g. "Narrated Abu Huraira:"). Banning it would create a structurally unfair comparison against the ground truth.

Step 3 — 03_evaluate.py

Every row is scored on two independent axes, because text-match and theme-match catch different kinds of failure:

- **Level 1 — BLEU score (text match):** is the model quoting the exact words of the real hadith? Computed with nltk's sentence_bleu with smoothing, scored 0 to 1.
- **Level 2 — Cosine similarity (theme match):** does the model's answer mean the same thing, even in different words? Both texts are embedded using the multilingual sentence-embedding model `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`, and cosine similarity between the two resulting vectors is computed, scored 0 to 1.

The embedding model runs **entirely locally** — it is downloaded once from Hugging Face and cached on disk; no network calls happen during scoring. One multilingual model is used for both English and Arabic, so cross-language comparisons stay consistent.

Each row is then classified into one of four labels:

Label	Condition	Meaning
correct	BLEU > 0.4 and cosine > 0.8	Model reproduced the real hadith closely
theme_only	BLEU < 0.4 and cosine > 0.8	Right general idea, wrong/paraphrased wording
hallucinated	cosine < 0.5	Model invented unrelated content
partial	everything else	Somewhere in between

Output: `results/{file}_evaluated.csv` with all original columns plus bleu_score, cosine_score, and label, along with a printed summary of label counts and average scores per book.

Step 4 — 04_results_summary.py

Combines every evaluated results file in results/ into one comparison table — average BLEU, average cosine, and the percentage breakdown of correct / theme_only / hallucinated / partial — grouped by model and language. Saves results/final_summary.csv and prints it to the terminal.

5. Understanding BLEU vs. Cosine Similarity

BLEU — "did it use the same words?"

BLEU compares overlapping word sequences (n-grams) between the model's answer and the real hadith text. It is a strict, literal measure: even a perfectly accurate paraphrase can score near zero if the exact wording differs. In this benchmark, low BLEU scores (typically 0.00–0.02 in the pilot run) mainly reflect that models are not reciting memorized hadith text verbatim — they are generating new text.

Cosine similarity — "did it mean the same thing?"

Both the model's response and the ground-truth hadith are converted into numeric vectors by the embedding model. Cosine similarity measures the angle between those two vectors: a small angle (score near 1) means the two texts occupy a similar region of "meaning space", regardless of exact wording. This lets the benchmark credit paraphrases and penalize genuinely unrelated content.

A caveat found during testing

For Arabic religious text specifically, cosine similarity can be misleadingly high (0.7–0.96 was observed) even when the hadith content is completely fabricated. This happens because formulaic phrases common to nearly every hadith — e.g. "■■■ ■■■■ ■■■■ ■■■■" (peace and blessings be upon him) or "■■■ ■■■■ ■■■■" (the Messenger of Allah said) — push the embeddings close together regardless of the actual narrated content. This is precisely why BLEU and cosine are used **together**, not in isolation: a hallucinated hadith can look thematically close on cosine alone while being entirely invented.

6. Models Evaluated (Pilot Run)

A small-scale pilot — 1 hadith sampled per book across all 6 books (6 hadiths total), run in both English and Arabic — was used to validate the full pipeline end-to-end before scaling up to the full 1000-hadiths-per-book sample. Three local Ollama models were tested in this version of the project:

- **mistral:7b**
- **deepseek-r1:8b**
- **llama3.1:8b**

Pilot results summary

Model	Lang	Avg BLEU	Avg Cosine	% Correct	% Theme Only	% Halluc.	% Partial
deepseek-r1:8b	arabic	0.011	0.652	0%	33.3%	33.3%	33.3%
deepseek-r1:8b	english	0.003	0.431	0%	0%	33.3%	66.7%
llama3.1:8b	arabic	0.015	0.478	0%	0%	50.0%	50.0%
llama3.1:8b	english	0.020	0.402	0%	0%	66.7%	33.3%
mistral:7b	arabic	0.008	0.725	0%	33.3%	16.7%	50.0%
mistral:7b	english	0.007	0.589	0%	0%	16.7%	83.3%

Observations

- **0% "correct" across every model and language** in this pilot — no model reproduced exact hadith wording from a bare reference (book / chapter / hadith number alone).
- **llama3.1:8b hallucinated the most**, especially in English (66.7% of pilot hadiths).
- **mistral:7b hallucinated the least** and had the highest average cosine score — though part of this is the formulaic-Arabic-phrasing effect described above, not necessarily better factual recall.
- **English hallucination rates were higher than Arabic** for deepseek-r1 and llama3.1, despite English generally having more training data available to these models — a pattern worth investigating further at full scale.
- All three models tend to produce **fluent, confidently-formatted fake hadiths** — correct narrator-citation style, and sometimes even fabricated source citations such as "[Sahih Bukhari 1.780]" — rather than declining to answer. This confident fabrication is the core hallucination risk this benchmark is built to surface and measure.

Note: this is a 6-hadith pilot intended to prove the pipeline works correctly end-to-end. The findings above are directional, not statistically robust. The full benchmark run — 1000 hadiths per book × 6 books = 6000 hadiths, per model, per language — is required for reliable per-book and per-model comparisons.

7. Requirements & Running the Pipeline

```
pandas | nltk | sentence-transformers | requests | tqdm | numpy
```

Install with `pip install -r requirements.txt`. Also requires Ollama running locally (`ollama serve`) with the target models pulled (e.g. `ollama pull mistral`).

```
python3 01_sample_data.py
python3 02_run_llm.py --model mistral:7b --lang english
python3 02_run_llm.py --model mistral:7b --lang arabic
python3 03_evaluate.py --file results/mistral:7b_english.csv
python3 03_evaluate.py --file results/mistral:7b_arabic.csv
python3 04_results_summary.py
```

Steps 2 and 3 are repeated for each additional model and language, then step 4 is run once at the end to combine every evaluated result into the final comparison table.